Image2Biomass Prediction with CEMS, CHARMS, and DA-Fusion

INF367A – CSIRO Kaggle Competition

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Abstract

We evaluate three recently published methods for small-data multi-target regression on the CSIRO Image2Biomass Kaggle competition, where the task is to predict five continuous biomass targets from 357 pasture photographs. Our baseline freezes a DINOv2 ViT-S/14 backbone and trains a lightweight MLP head with a per-target weighted Smooth-L1 loss. On top of this baseline, each team member adapts one method targeting a different hypothesised bottleneck: **DA-Fusion** [Trabucco et al., 2024] generates diffusion-based synthetic training images; **CEMS** augments training along the local tangent space of the joint feature-target manifold; and **CHARMS** aligns image features with auxiliary tabular attributes available at training time. We analyse which methods compose natively on a shared backbone and build a combined pipeline integrating all three methods. Individually, all three methods perform near the baseline; CEMS narrowly edges ahead on the private leaderboard (+0.002 R^2) while the other two fall slightly short. The combined pipeline underperforms all individual methods. We discuss why each method's core assumptions interact poorly with continuous targets, strong pretrained features, and a training set of only a few hundred images.

1 Introduction

Estimating pasture biomass from imagery supports grazing management and yield forecasting, but is characterised by small labelled datasets, fine visual distinctions between species, and continuous multi-target labels rather than discrete classes—properties that limit what standard vision pipelines deliver out of the box.

Self-supervised vision transformers such as DINOv2 [Oquab et al., 2023] offer a natural starting point: pretrained features transfer well with limited data, and freezing the backbone mitigates overfitting on small training sets. A frozen DINOv2 ViT-S/14 with a lightweight MLP head forms our baseline. On top of this baseline, each group member contributed one recently published method addressing a different hypothesised bottleneck: DA-Fusion [Trabucco et al., 2024] targets data scarcity via diffusion-based synthetic augmentation; CEMS targets geometric structure in the joint feature-target manifold; CHARMS [Jiang et al., 2024] targets the use of auxiliary tabular signal available at training but not at test time.

Our contributions are:

(1) implementations and evaluations of each method under a unified baseline;

(2) an integration analysis identifying which methods compose;

(3) a combined pipeline integrating all three methods on a shared backbone; and

(4) a discussion of where each method's assumptions hold or break in the continuous-target small-data regime.

2 Problem and Data

The CSIRO Biomass Kaggle competition requires predicting five continuous biomass targets from a single RGB pasture photograph: dry weights of green biomass, dead biomass, clover, GDM, and total biomass.

The metric is a weighted global R^2, computed by flattening all (image, target) pairs into one vector with per-target weights 0.1/0.1/0.1/0.2/0.5; the *Total* target therefore carries half the weight.

The training set contains 357 unique pasture photographs at 2000×1000 resolution. The small size, visual similarity across species, and continuous targets make this a challenging small-data regression problem. We use `GroupKFold` cross-validation with five folds, grouping by source so all orientation-augmented copies fall in the same fold; this prevents leakage introduced by flip-based augmentation. Predictions are clipped at zero since biomass cannot be negative.

Each training image is expanded to four orientations (identity, horizontal flip, vertical flip, 180°), yielding 1428 training rows. This augmentation is shared across all methods.

3 Baseline

Our baseline is a frozen self-supervised feature extractor with a lightweight trainable head, chosen to make the most of a small training set without overfitting. Images are resized to 504×252, normalised with ImageNet statistics, and passed through a frozen DINOv2 ViT-S/14 backbone [Oquab et al., 2023]. We use the 384-d `[CLS]` token as the image representation; the positional embeddings are interpolated to the non-square input resolution.

The head, *BiomassModel*, is a small MLP with an encoder ($384 \to 256 \to 64$) followed by a regression head ($64 \to 32 \to 5$). Each linear layer is followed by GELU activation and dropout of 0.3. Only the head is trained, yielding roughly 117k trainable parameters.

We train with a per-target weighted Smooth-L1 loss using the competition weights ({Green, Dead, Clover}: 0.1, GDM: 0.2, Total: 0.5), optimised with AdamW at learning rate 3×10^{-4}, weight decay 10^{-3}, batch size 64, for 80 epochs. Learning rate follows a cosine schedule with $\eta_{\min} = \eta_0/100$. Best checkpoints per fold are selected by validation weighted R^2. All three method sections below modify this baseline in one component at a time; other components remain unchanged unless explicitly stated.

4 Methods

4.1 CEMS

Motivation. With only 357 labelled images, the training distribution is sparse and standard ERM trains on a fixed set of feature-target pairs. The manifold hypothesis suggests that DINOv2 features and continuous biomass targets lie near a low-dimensional manifold; sampling *along* that manifold should yield more meaningful synthetic points than naive interpolation. CEMS [Kaufman & Azencot, 2025] operationalises this with a second-order local approximation that accounts for the curvature missed by first-order methods.

Adaptation. We operate in the *ambient* joint space $\mathbf{z}=[\mathbf{x}, \mathbf{y}]$, where $\mathbf{x}\in\mathbb{R}^{384}$ is the frozen DINOv2 CLS token and $\mathbf{y}\in\mathbb{R}^5$ are the biomass targets, giving a 389-d joint space. Because targets span very different scales (0–186 g), we MinMax-scale $\mathbf{y}$ to $[0, 1]$ before forming the joint space and inverse-transform synthetic labels back to raw scale for the loss.

Pipeline. For each minibatch of size B=64, CEMS centres the batch, computes a local SVD basis B_u, then estimates the gradient ∇g and Hessian $\mathbf{H}$ of the normal-space embedding via ridge regression on projected neighbour differences. A perturbation $\boldsymbol{\eta}\sim\mathcal{N}(\mathbf{0}, \sigma\mathbf{I}_d)$ is drawn in the d-dimensional tangent space and mapped back via

$$\tilde{\mathbf{z}} = B_u\big[\boldsymbol{\eta},\ \boldsymbol{\eta}^\top\nabla g + \tfrac{1}{2}\boldsymbol{\eta}^\top\mathbf{H}\,\boldsymbol{\eta}\big] + \boldsymbol{\mu}_{N_z}. \tag{1}$$

The intrinsic dimension d is re-estimated per batch with TwoNN [Facco et al., 2017]; in practice d=2 throughout training. We use fixed σ=1 and train on CEMS-synthesised samples only (`mix_real=False`).

Results. Table 1 reports validation and Kaggle performance.

Table 1: Validation and Kaggle test weighted R^2. Higher is better.

Model	Val R^2	Kaggle public	Kaggle private
Baseline	0.7554	0.5533	0.5090
CEMS	0.7668	0.5497	0.5106
Δ	+0.0114	−0.0036	+0.0016

Discussion. CEMS is the only individual method that does not degrade private leaderboard performance, gaining +0.002 R^2 over the baseline, though this margin is within run-to-run noise. The estimated intrinsic dimension of the joint space is d= 2.19, far below the 389-d ambient dimension, indicating that the frozen DINOv2 backbone already captures most meaningful variation. Sampling within this two-dimensional manifold introduces only mild diversity, limiting the benefit of the second-order curvature correction.

4.2 CHARMS

Motivation. The CSIRO Biomass dataset includes expert-measured attributes available at training time—pasture height, GreenSeeker NDVI, geographic state, and dominant species—but not at inference. This matches the setting CHARMS [Jiang et al., 2024] is designed for: leveraging tabular side-channel information during training to improve an image-only model at test time.

Adaptation. The original method operates on CNN spatial feature maps; here we use the 384-dimensional CLS token from a frozen DINOv2 ViT-S/14, treating each dimension as a channel. k-Means clustering ($k = 40$) groups dimensions before the OT cost matrix is computed. Four attributes are used: two numerical (`Pre_GSHH_NDVI`, `Height_Ave_cm`) and two categorical (`State`, `Species_first`), giving $\mathbf{T} \in \mathbb{R}^{4\times 384}$.

Pipeline. An FT-Transformer [Gorishniy et al., 2021] encodes tabular attributes into per-attribute embeddings and produces an auxiliary biomass prediction. An Optimal Transport step aligns image dimensions with attributes via inter-sample cosine similarity, yielding transfer matrix $\mathbf{T}$. Li2t heads then predict each attribute from $\mathbf{T}$-weighted image features. The total loss is:

$$\mathcal{L} = \mathcal{L}_{\text{image}} + \lambda_{\text{tab}}\,\mathcal{L}_{\text{tabular}} + \lambda_{\text{li2t}}\,\mathcal{L}_{\text{i2t}} \tag{2}$$

with $\lambda_{\text{tab}} = 0.5$, $\lambda_{\text{li2t}} = 0.1$, and $\mathbf{T}$ recomputed every five epochs.

Results. Table 2 and Figure 1 report validation and Kaggle performance over 80 epochs.

Table 2: Validation and Kaggle test weighted R^2. Higher is better.

Model	Val R^2	Kaggle public	Kaggle private
Baseline	0.7935	0.5533	0.5090
CHARMS	0.7774	0.5435	0.5080
Δ	−0.0161	−0.0098	−0.0010

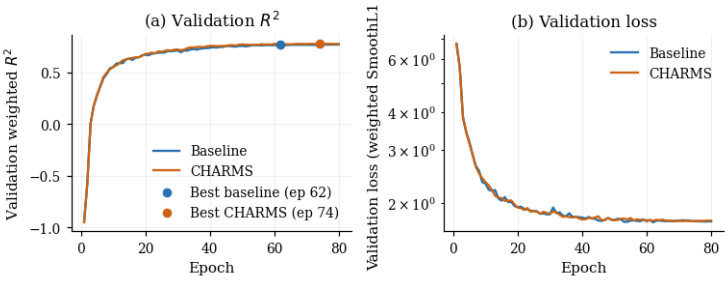

Figure 1: Validation curves over 80 epochs. CHARMS peaks at epoch 74 ($R^2 = 0.7774$) versus epoch 62 for the baseline ($R^2 = 0.7697$), with similar convergence in both R^2 and loss.

Discussion. CHARMS does not improve over the baseline. The CLS token is lower-dimensional than the CNN feature maps the method was designed for, limiting the quality of the OT alignment. The dataset is also far smaller than those in the original paper, making similarity estimates noisy. Finally, the regression setting is a mismatch: the original paper targets classification, where discrete labels provide stronger supervision for the auxiliary Li2t losses.

4.3 DA-Fusion

Motivation. DA-Fusion [Trabucco et al., 2024] expands small training sets by generating synthetic images with Stable Diffusion, using textual inversion [Gal et al., 2023] to learn a token per fine-grained class. It was proposed for image classification, where each synthetic inherits its source image's class label. We adapt it to our 5-target biomass regression setting, where labels are continuous dry-weight vectors.

Adaptation. Textual inversion assumes discrete categories, so we use pasture species as the grouping: one `<species_token>` per species with at least 10 images, and a generic prompt otherwise. The label-copy rule is stronger for regression than for classification, since continuous targets are sensitive to visual changes that a class label would absorb. We mitigate this by keeping img2img strength low (0.2–0.6, jittered) so synthetics remain visually close to their source, and apply a per-fold leakage filter excluding synthetics whose source image is in the validation split.

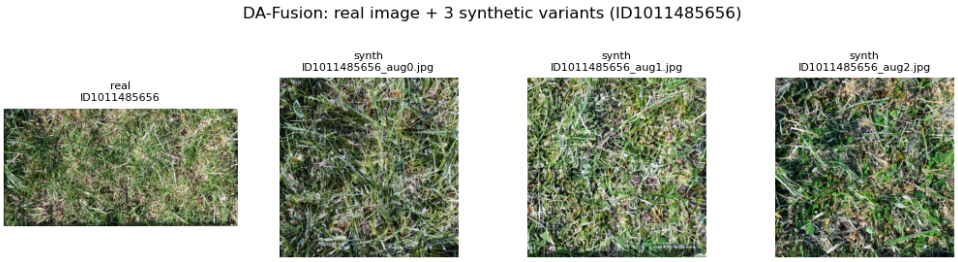

Figure 2: One real image (left) and three DA-Fusion synthetics.

Pipeline. Textual inversion and image synthesis are run offline, producing 1071 synthetic images paired with their source image's target vector. At training time, synthetics are injected into the DINOv2 + MLP baseline; all other components (backbone, head, loss, optimiser, schedule) are unchanged.

Results. On the Kaggle leaderboard, DA-Fusion with the DINOv2 backbone scored 0.531 public / 0.498 private weighted R^2, compared to the baseline's 0.532 / 0.509. A ResNet-50 variant, tried as a backbone ablation, scored 0.483 / 0.425. To disentangle the effect from classical pixel-space augmentation, we ran a local fold-0 ablation against a RandAugment baseline under the identical DINOv2 + MLP pipeline: RandAugment lifts validation R^2 by +0.11, whereas DA-Fusion lifts it by only +0.07 and has strictly worse per-target RMSE on Green, Clover, and Total. Both augmented runs also converge substantially faster than the plain baseline, reaching near-peak validation R^2 within $\sim$15 epochs versus $\sim$40 for the baseline. The Kaggle and local trends agree: DA-Fusion helps less than simple classical augmentation.

Discussion. Three factors likely contribute. **(1)** The label-copy assumption is weaker in regression than in classification: a Stable Diffusion edit that would leave a class label intact can shift continuous biomass targets, injecting label noise. **(2)** The DINOv2 backbone is pretrained at a scale that may already saturate the signal extractable from 357 training images, leaving little room for *added* data to help; RandAugment still wins because it regularises the head via feature-level perturbation rather than adding new samples. **(3)** Synthetics outnumber real images after injection and may dilute real-image gradients, an effect RandAugment avoids since it augments the real images in place. Continuous-target tasks may need a label-aware generation step (e.g. conditioning on the target vector) to avoid the label-noise issue.

5 Combined Pipeline

The assignment asks us to combine our individual methods into a single system. Before implementation, we analysed which of the three methods compose natively on our shared DINOv2 + MLP baseline.

Compatibility analysis. DA-Fusion and CEMS are both training-time modifications to a frozen DINOv2 backbone and compose cleanly: DA-Fusion expands the training set, and CEMS then builds its k-NN graph on the expanded $[\mathbf{X}, \mathbf{Y}]$ space. The only ordering constraint is that synthetics must be injected *before* the k-NN graph is built, and the target scaler must be fitted on the combined training labels.

CHARMS is also compatible: treating the 384 CLS dimensions as channels, as described in Section 4.2, allows the OT alignment to operate on the same frozen DINOv2 features used by the other two methods.

Full CHARMS integration. The combined pipeline runs the complete CHARMS stack on the 384-d CLS token. An FT-Transformer encodes four tabular attributes (NDVI, pasture height, state, species) and produces an auxiliary biomass prediction; K-Means (k=40) clusters image dimensions by co-activation; Earth Mover Distance aligns channel- and attribute-similarity matrices to yield $\mathbf{T} \in \mathbb{R}^{4\times384}$, recomputed every five epochs. CHARMS losses are applied only to real samples, as synthetics carry no tabular data.

Combined training loop. For each `GroupKFold` split: (i) filter synthetics to the training split; (ii) extract DINOv2 features and tabular attributes for real training rows; (iii) fit the target scaler on the combined labels; (iv) cluster 384 image dimensions via K-Means and initialise $\mathbf{T}$ uniformly; (v) train with the CEMS anchor loop—CHARMS losses computed on the real-sample subset of each anchor batch, $\mathbf{T}$ refreshed via EMD every five epochs. Checkpoint selection by validation weighted R^2.

5.1 Results

Table 3 reports the combined pipeline's performance on the Kaggle leaderboard alongside the baseline and each individual method for reference.

Pipeline	Public R^2_w	Private R^2_w
Baseline (DINOv2 + MLP)	**0.553**	0.509
DA-Fusion	0.531	0.498
CEMS	0.549	**0.510**
CHARMS	0.543	0.508
Combined (DA-Fusion + CEMS + CHARMS)	0.549	0.485

Table 3: Weighted R^2 on the Kaggle leaderboard. The combined pipeline does not recover baseline performance.

The combined pipeline scores 0.549/0.485 public/private weighted R^2—matching CEMS on the public leaderboard but falling below all individual methods on the private score. Combining the three modifications does not recover baseline performance.

5.2 Discussion

The combined pipeline confirms the pattern visible in the individual results: each method's modification reduces performance, and stacking them compounds rather than cancels the effect. This is consistent with each method addressing a *hypothesised* bottleneck that turns out not to be the real one in this setting—data scarcity for DA-Fusion, latent-space geometry for CEMS, and cross-modal supervision for CHARMS. When the actual constraint is elsewhere (plausibly the quality of the 5-target label signal given 357 images and a frozen backbone that already saturates available representation capacity), adding components that each introduce their own noise or regularisation pressure makes the system worse, not better.

Limitations. Our conclusions come with caveats. Each method is evaluated with the hyperparameters reported in its source paper plus minimal tuning; a more thorough search might reduce the gap to baseline. Kaggle leaderboard scores reflect a single training run per method rather than a seed-averaged mean, so differences of ±0.01 R^2 are within run-to-run noise.

6 Conclusion

We adapted three recently published methods to a 5-target pasture-biomass regression task with 357 training images and a frozen DINOv2 backbone. Individually, all three methods perform near the baseline; CEMS narrowly edges it out on the private leaderboard (+0.002 R^2) while DA-Fusion and CHARMS fall slightly short. The combined pipeline underperforms all individual methods. We attribute this to mismatches between each method's target domain (classification, geometric regularity, or cross-modal supervision) and the actual constraints of our setting, and suggest that continuous-target small-data regression problems may benefit more from label-aware generation and head-level regularisation than from the forms of augmentation tested here.

References

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